

◆ NIRAJ NAIDU

— MECHANICAL PRODUCT DESIGNER • CAD ENGINEER •

NIRAJ NAIDU

MECHANICAL PRODUCT DESIGNER • HYDRAULIC SYSTEMS •

A detail-oriented mechanical product designer with hands-on experience in 3D modeling, hydraulic system design, reverse engineering, and precision machining — delivering solutions from gantry cranes and mold presses to tube-bending machines and Formula Student race cars.

MECHANICAL DESIGN ENGINEER

ESSEN, GERMANY

FIG. 01 - PROFILE



Location	Essen, 45138
DOB	11.04.2001
Phone	+49 15757055980
Email	nirajnaidu0411@gmail.com
Focus	Additive Mfg.
Status	M.Sc. Student

— 01 / ABOUT

ENGINEERING WITH PRECISION

Currently pursuing a Master of Science in Digital Technologies at FOM University of Applied Sciences for Economics and Management, Germany, with a research focus comparing the sustainability of CNC machining against metal additive manufacturing.

Previously earned a B.Tech in Mechanical Engineering from Symbiosis Institute of Technology, Pune, where hands-on project work included leading the design of Formula Student race car components and hydraulic systems for industrial applications.

Comfortable moving between CAD platforms — SOLIDWORKS, CATIA and AutoCAD — and validating designs in ANSYS, with practical Additive Manufacturing experience spanning Cura and RD Works workflows. Tolerances held as tight as 5 microns on production fixture work.

EN — English (C1)

HI — Hindi (B1)

MR — Marathi (B1)

DE — German (A1)

— 02 / TOOLSET

SOFTWARE & CAPABILITY

SOLIDWORKS 95%

AutoCAD 88%

Additive Manufacturing 90%

CATIA 85%

ANSYS 78%

Cura / RD Works 82%

PROJECT & WORK HISTORY

JAN 2024 – PRESENT

Freelance

ADDITIVE MANUFACTURING PRODUCT & DESIGN

- Deliver 3D modeling and 3D-printing solutions for clients, prioritizing accuracy and design efficiency.
- Develop prototypes and production-ready CAD models tailored to client specifications.
- Provide design consulting focused on precision, material selection, and assembly fit.

JUN 2024 – DEC 2024

KC Engineering and Automation

DESIGN ENGINEER

- Designed and modeled hydraulic systems, including cylinders, presses, and lifting equipment, using SOLIDWORKS.
- Directed the reverse engineering of the SLB 1100 Gantry Crane to improve structural efficiency and manufacturing feasibility.
- Designed and developed hydraulic tube-bending machines, mold-press equipment, and horizontal moving jacks.
- Created detailed 3D CAD models and engineering drawings, ensuring precision, manufacturability, and assembly accuracy.
- Prepared manufacturing documentation and collaborated with the fabrication team to support production and assembly.

JAN 2024 – MAY 2024

ASB International Pvt. Ltd.

DESIGN INTERN

- Created 3D models of mold components and designed precision fixtures for industrial machinery.
- Maintained tolerances of 5 microns to ensure assembly stability during machining.
- Supported the design team in drafting engineering documentation and production drawings.

2022 – 2023

Wrench Welders Racing – Formula Student Team, SIT

DESIGN HEAD

- Led the design and fabrication of Formula Student race car components using SOLIDWORKS and CATIA.
- Designed and manufactured carbon-fibre and fibreglass body panels for improved vehicle performance.
- Developed a high-efficiency EV battery pack, contributing to a 15th-place finish among 72 competing teams.
- Built an engine sound signature segregation system using audio analysis and machine learning for vehicle diagnostics.
- Represented the team at SUPRA 2022 and Formula Student EV 2021.

ACADEMIC RECORD

APR. 2025 – APR. 2027

Master of Science in Digital Technologies

FOM Hochschule für Oekonomie & Management, Germany

JUN. 2020 – JUN. 2024

Bachelor of Technology in Mechanical Engineering

Symbiosis Institute of Technology, Pune — CGPA 7.0 / 10

FEB. 2018 – FEB. 2019

Higher Secondary Certificate, Science

Fr. Agnel's Multipurpose School and Jr. College — 70.77%

MAR. 2016 – MAR. 2017

Secondary School Certificate Standard

Fr. Agnel's Multipurpose School and Jr. College — 75.00%

LET'S BUILD SOMETHING PRECISE

Open to product design roles, freelance CAD work and collaboration on hydraulic systems or additive manufacturing projects.

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CV DESIGNED TO MATCH PERSONAL PORTFOLIO WEBSITE